

# The Tulip Mania and the Psychology of Speculative Bubbles

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## 1 Introduction

Throughout history, financial markets have experienced particular episodes in which asset prices rose far beyond their intrinsic value only to collapse abruptly. One of the best examples is The Tulip Mania of the 1630s. It took place in the Netherlands and is often perceived as the first speculative bubble to be documented in history. At the peak of this phenomenon, the price of tulip bulbs rose to extraordinary levels, in some cases exceeding the value of a house. Eventually, when the market collapsed, thousands of investors faced severe financial losses. This phenomenon exemplifies a speculative bubble. However, from a psychological perspective, it reflects a complex interaction of emotions, cognition, and social influence that drive collective irrationality.

## 2 Psychological Mechanisms Behind Speculative Bubbles

The Tulip Mania offers a classic framework for understanding the psychology of speculative bubbles. Many emotional and cognitive mechanisms support collective irrationality.

**Herd Behavior** - it is said to be one of the strongest forces in speculative bubbles and is defined as a tendency to follow the crowd under the assumption that others possess superior knowledge. Banerjee (1992) described this phenomenon as *informational cascades*, where individuals ignore their private information and imitate others' actions. During Tulip Mania, people bought tulip bulbs because others were doing so, believing it to be a rational decision.

**Fear of Missing Out (FOMO)** - is believed to drive people to act impulsively to avoid feeling left out. Shiller (2000) noted that this psychological factor often amplifies bubbles, as individuals join the market not based on fundamentals, but on the anxiety of missing profits that others are achieving.

Even the most skeptical observers eventually succumbed to tulip speculation once they saw others making money.

**Overconfidence Bias** - leads individuals to overestimate their predictive abilities and control over outcomes. Odean (1998) demonstrated that overconfident investors trade excessively, believing that they can time the market. Many tulip sellers were convinced that they could sell before the inevitable crash, only to be caught by surprise when the prices collapsed.

**Emotional Contagion** - is another important factor where emotions spread rapidly through social networks. Hatfield, Cacioppo, and Rapson (1993) explained that shared emotional states, such as excitement or euphoria, can synchronize across groups, fueling collective enthusiasm. The social exuberance surrounding the trade of tulips turned speculative behavior into a cultural movement.

**Cognitive Dissonance** - is used to explain why people remain committed even when warning signs appear. Festinger (1957) proposed that people seek consistency between beliefs and actions; therefore, once investors committed the money, they justified their behavior by convincing themselves that the tulips were genuinely valuable. Their self-justification reduced psychological discomfort but increased financial risk.

### 3 Social Media Analysis: Modern Echoes of Speculative Bubbles - GameStop Short Squeeze

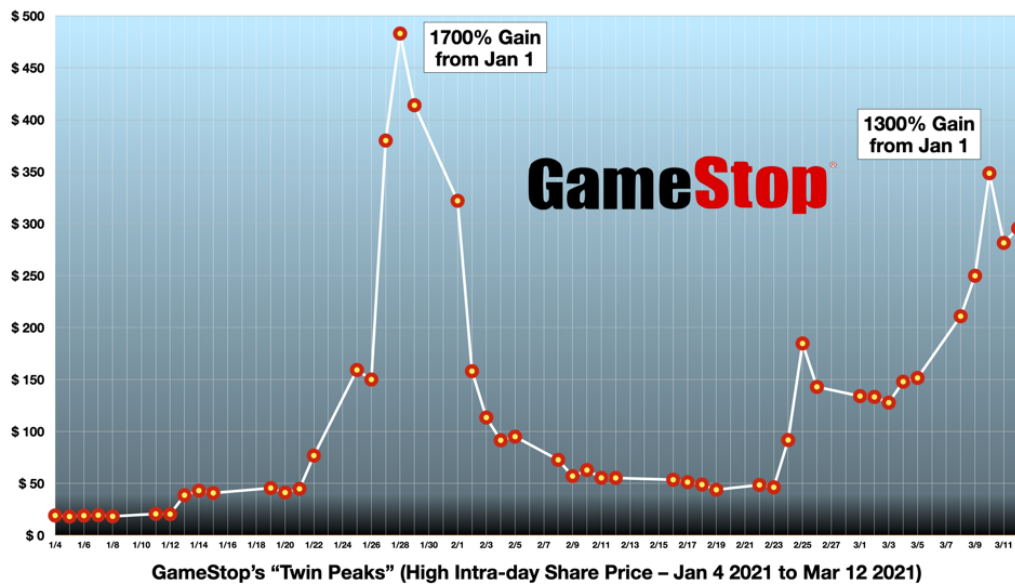


Figure 1: GME'S Twin Peaks – January and February Surges. Chart by George Calhoun

In 2021, retail investors on Reddit's *r/WallStreetBets* collectively bought shares of GameStop (GME), a heavily shorted stock, as a symbol of rebellion against institutional hedge funds. Through viral memes and social media posts, millions of small investors bought shares and call options, driving the price of GME from under \$20 to \$480 in a few days (Figure 1).

While the Tulip Mania unfolded in public markets, GameStop's bubble evolved in the digital world. Nevertheless, the underlying psychology remained identical.

Psychological parallels:

- **Herd behavior & groupthink** - thousands followed the crowd without independent thinking and analysis.

- **Emotional Contagion** - memes and collective excitement fueled a sense of shared victory.
- **FOMO** - many joined just to avoid "missing out" on seemingly easy profits.
- **Cognitive Dissonance** - investors justified holding even as prices crashed (*diamond hands*).

Calhoun (2021), in his analysis of the situation, argued that GameStop's surge was not merely a *short squeeze*, when short-sellers are forced to buy back shares, but also a *gamma squeeze*. Such squeeze occurs in the option market, particularly when traders buy large quantities of call options, the institutions which sell them must hedge their exposure by buying the underlying stock. As the stock price rises, said hedgers are forced to buy even more shares, thus creating a self-reinforcing cycle of demand. This dynamic, initially driven by the enthusiasm of the crowd, transformed into a feedback loop of automatic buying by professional market makers.

This interaction between psychology and market structure highlights how modern technology amplifies behaviors from the past and how hard it can be for humans to learn from the mistakes of others. In the 1630s, human exuberance was enough to inflate the prices and in 2021, the same exuberance triggered powerful algorithmic and derivative mechanisms that magnified the scale of the situation exponentially.

## 4 Conclusion

The Tulip Mania illustrates that economic irrationality often stems not from a lack of intelligence, but from deeply human psychological biases. Emotions, social pressures and self-justifying thought patterns create fertile conditions for bubbles to form and burst. Understanding these mechanisms is crucial not only for economists but also for psychologists seeking to comprehend how

individual cognition aggregates into collective irrationality. Both in the 17th century and the current times, the same mental processes - herd behavior, overconfidence, FOMO, emotional contagion and cognitive dissonance, continue to shape human economic behavior. Moreover, understanding these parallels matters because the next bubble, whether in cryptocurrencies or another asset class, will most likely be driven by the same timeless forces. The analyses show that the mechanics may change, but the mind of the market remains the same.

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